

A Bayesian Exploration on the Motivational and Behavioral Impacts of Chatbots in Language Learning

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Abstract

This study investigates the motivational and behavioral effects of chatbot interfaces on knowledge tasks in language learning, focusing on English as a Second Language (ESL) learners. As chatbots gain prominence in education, understanding how user factors—metacognitive awareness, help-seeking behavior, and technology acceptance—interact with interface design is critical for optimizing engagement and performance. We conducted an experimental study comparing conversational and informational chatbots during audio-based learning tasks. Bayesian analysis revealed that chatbot type minimally influenced perceived usefulness, with informational chatbots showing slightly higher engagement. Metacognitive awareness strongly predicted help-seeking tendencies, which positively impacted technology acceptance. These findings offer actionable insights for designing AI systems that enhance user engagement and learning outcomes while balancing reliance on automation. Our work contributes to understanding human-AI interaction in educational technology, emphasizing user-centered design's role in fostering equitable and effective learning experiences.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; **Natural language interfaces**; • **Social and professional topics** → **User characteristics**.

Keywords

chatbot, conversational interfaces, human-AI collaboration, educational technology, English as second language

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1 Introduction

Large language models (LLMs), such as ChatGPT, have rapidly transformed the landscape of artificial intelligence, finding applications in fields as diverse as scientific research, creative writing, healthcare, legal analysis, and education [1, 18, 29]. This research focuses on how chatbot interfaces can support English as a Second Language (ESL) learners by examining the interplay between user engagement, help-seeking behavior, and metacognitive awareness. The ultimate goal is to provide actionable insights for designing AI systems that enhance learning outcomes and promote equitable access.

In education, LLMs have proven effective in supporting open-internet exams, providing real-time feedback, and enhancing language learning [13, 17, 32]. For ESL learners, conversational AI tools address key challenges such as limited access to native speakers and the need for personalized instruction [8]. These tools offer interactive platforms for language practice, enabling learners to improve their skills in authentic contexts.

However, the use of LLMs is not without limitations. Known issues include the generation of “hallucinations” and difficulties in understanding domain-specific contexts, which can undermine trust in AI [10, 19]. Additionally, excessive reliance on these tools may lead to reduced critical thinking and skill degradation, as noted in studies on cognitive offloading [11, 14, 15]. These challenges underscore the importance of designing AI systems that foster appropriate reliance and user trust [6].

Despite these limitations, dismissing the use of AI entirely would overlook its potential to empower learners and enhance educational outcomes. Research shows that AI systems can help users achieve aspirations [23] which can be linked to improved well-being and performance [12, 27]. When responsibly designed, these tools complement human cognition, enabling learners to perform at their best [7, 20].

Finally, user factors play a critical role in determining the effectiveness of AI-based interventions. Individual differences in metacognitive awareness, self-efficacy, and help-seeking behavior significantly influence how users engage with AI tools. Research on technology acceptance has shown that users' attitudes toward AI shape their interactions with LLMs, which power many contemporary chatbots [2]. Help-seeking behaviors, which drive the decision to use tools like AI, can be either beneficial or maladaptive depending on the context [22]. In language learning, metacognitive awareness is crucial in determining when and how to seek help,

Table 1: Number of participants who identified English proficiency levels based on Common European Framework of Reference for Languages

CEFR Level	Description	Reading	Writing	Listening	Speaking
A1	Use simple phrases for basic needs at best	0	0	0	0
A2	Use the language for everyday activities	0	0	0	0
B1	Use for simple conversations about familiar topics	1	0	0	2
B2	Use confidently for many topics	6	6	12	8
C1	Use fluently in any situation	12	13	10	12
C2	Use with complete mastery	11	11	8	8

a process that is essential for optimizing AI tool usage [30]. Understanding these factors is key to designing AI systems that are both effective and equitable, ensuring that AI tools meet the diverse needs of users.

We investigate the effects of conversational and informational chatbot interfaces on ESL learners engaged in audio-based learning tasks. The audio format was chosen to simulate real-world scenarios, such as note-taking during lectures while encouraging interaction with chatbots by limiting reliance on text-based searches. Conversational interfaces are hypothesized to foster greater engagement and improved learning outcomes compared to informational interfaces. Factors such as perceived usefulness, engagement, technology acceptance, metacognitive awareness, and help-seeking behavior are expected to influence performance. Additionally, individual differences, including topic familiarity, interest, confidence, and English competency, are anticipated to moderate these effects. This work contributes to understanding how usage and metacognitive factors shape learning outcomes and inform the design of AI systems tailored to diverse educational needs.

2 Methods

The sample consisted of 30 adults from Prolific, an online participant pool. The mean age was 35.5 years ($SD = 11.9$, range: 21–69). Of the participants, 10 identified as female and 20 as male. They came from 17 different nationalities at birth. The inclusion criteria required them to be non-native English speakers but proficient in English, and equipped with a computer capable of audio playback. The self-reported English proficiency levels are shown in Table 1. Participants were compensated at Prolific’s recommended rates. All participants provided informed consent.

A custom web application guided participants through the experiment. The workflow involved:

- **Consent and Demographics:** Participants provided consent and completed a demographic questionnaire.
- **Briefing and Pre-experiment Questionnaires:** Participants read a document explaining the motivation behind the experiment and how the tasks would proceed. They completed a modified version of the Metacognitive Awareness Listening Questionnaire [30], adapted from its original French-learning context. Additional help-seeking questions were included, derived from Computer Science Help-Seeking Scales [22].

- **Onboarding Tutorial:** Participants were introduced to the study task (Fig. 1), which involved studying an audio recording and answering a quiz. An onboarding tutorial guided them through the screen’s features. They proceeded to the quiz after completing the tutorial and the accompanying questionnaire. Quiz results were shown upon submission or timeout.
- **Study Sessions:** After the onboarding tutorial, participants completed subsequent study sessions independently. The behavior of the note-taking area varied based on the assigned treatment condition. Study tasks were untimed, but quizzes were limited to 200 seconds. Each quiz contained eight questions modeled after the TOEFL iBT Reading section [28]. Participants completed three study sessions, including the one with the onboarding tutorial. Interaction logs (e.g., clicks, timestamps) were recorded for analysis.
- **Post-experiment Questionnaires:** Participants completed a final questionnaire, including the User Engagement Scale Short Form [21] adapted for chatbots and a general questionnaire on large language model acceptance [2].

This between-subjects experiment included two conditions dictating the behavior of the note-taking chatbot:

- **Informational chatbot:** Term Frequency-Inverse Document Frequency (TF-IDF) was used to identify the portion of the audio script most relevant to the participant’s question. TF-IDF is a statistical technique that reflects the importance of a term in a document relative to a collection of documents, based on its frequency and specificity [26]. Tokenization was performed using the Natural Language Toolkit (NLTK) [3], and TF-IDF vectorization was implemented using scikit-learn [24].
- **Conversational chatbot:** Information retrieved through TF-IDF was passed to a pre-trained FLAN-T5-small model. FLAN-T5 is a transformer-based language model fine-tuned on instruction-based learning tasks [33]. The model underwent three additional iterations of fine-tuning on question-answer pairs derived from the audio scripts, using the Hugging Face Transformers library [34]. Each iteration included training for 5 epochs with a batch size of 1 per device, gradient accumulation over 8 steps, and evaluation every 500 steps. The Seq2SeqTrainingArguments class was used to configure the fine-tuning process.

Bayesian methods were used to estimate parameters in small samples and to account for uncertainty. Priors included a normal

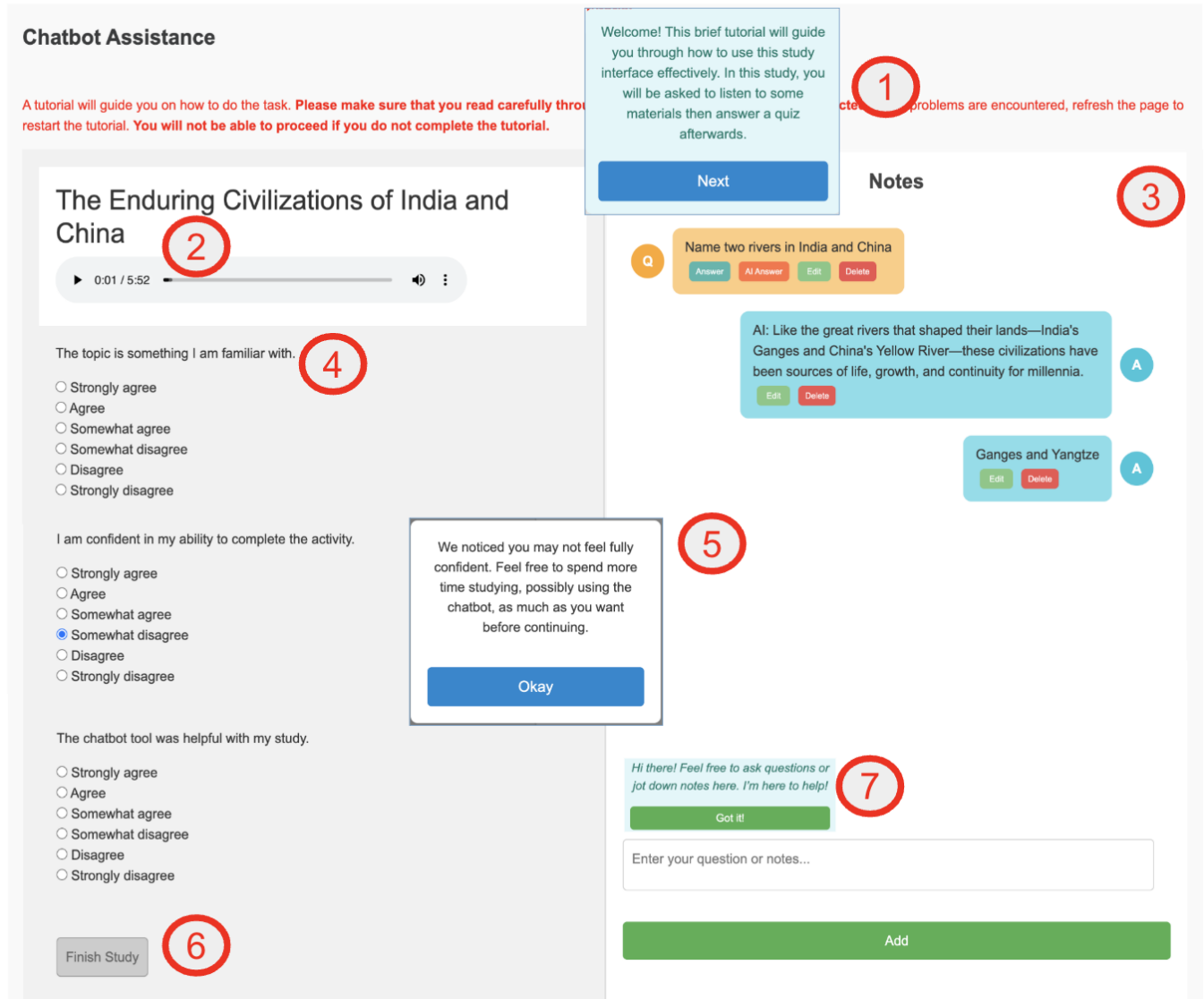


Figure 1: The user interface of the study task. On first access to the screen, the participants were guided through an onboarding tutorial to familiarize themselves with all parts of the screen (1). The screen included audio playback controls (2), a note-taking interface stylized as a chatbot (3), and a brief questionnaire requiring the participants to share their thoughts on the activity (4). Participants were encouraged to continue studying if they indicated low confidence in answering the associated quiz (5). They could click the [Finish Study] button after completing the questionnaire (6). On subsequent accesses, participants were prompted to use the note-taking feature (7).

prior ($N(0, 2)$) for regression coefficients and a Student- $t(3, 0, 10)$ prior for intercepts. Models were implemented in R using the brms package [4], leveraging the No-U-Turn Sampler (NUTS) [16] via the Stan backend [5]. Analyses included cumulative models for ordinal regression and Gaussian models for continuous variables. Each model used four chains, running on four cores, with 2,000 iterations per chain.

3 Results and Discussion

The internal consistency of all scales used was evaluated using Cronbach's alpha [9] (Table 2). The reliabilities ranged from $\alpha = 0.78$ to $\alpha = 0.88$, suggesting that the items consistently measure the intended constructs. The analysis was conducted using the psych package in R [25].

We began by investigating the influence of chatbot interface type (Informational vs. Conversational) on perceived usefulness and user engagement. Usefulness ratings were derived from participant responses in the questionnaires during each study session

Table 2: Reliability test results

Questionnaire	Number of Items	Cronbach α
Metacognitive Awareness Listening Questionnaire	21	0.78
Help-Seeking Questionnaire	11	0.88
User Engagement Scale (Short Form)	12	0.85
Technology Acceptance Questionnaire	35	0.88

about the chatbot's helpfulness. Bayesian cumulative ordinal regression was used to model these ratings, with weakly informative priors specified (normal prior, $N(0, 2)$, for regression coefficients and Student- $t(3, 0, 10)$ prior for intercepts). Posterior predictive checks confirmed a good model fit. As shown in Fig. 2 [1], participants rated the informational chatbot slightly higher in perceived usefulness than the conversational chatbot, with stronger agreement observed for statement "The chatbot was helpful with my study." For instance, the probability of a "Strongly Agree" rating was approximately 0.12 for the informational chatbot compared to 0.10 for the conversational chatbot. However, the posterior probability of 70% favoring the informational chatbot suggests only moderate evidence, with small effect sizes limiting practical significance.

Next, we examined engagement scores, which were computed as the average of participant responses to the User Engagement Scale (Short Form) [21], administered post-experiment. Bayesian linear regression was applied to these scores to explore whether chatbot type influenced engagement. The results, shown in Fig. 2 [2], indicated slightly higher predicted engagement for the informational chatbot (mean = 2.73) compared to the conversational chatbot (mean = 2.49). Although the credible intervals overlapped, suggesting no statistically significant difference, the trend aligns with the usefulness results, indicating that the informational chatbot's straightforward interaction style may have had a marginal positive influence on both perceived usefulness and engagement.

To explore the relationship between perceived usefulness and engagement, Bayesian hierarchical regression was used. Engagement scores were again modeled as the average of participant responses to the User Engagement Scale, while usefulness ratings from the study session questionnaire served as predictors. Fig. 2 [3] shows a flat relationship, with no significant change in predicted engagement scores across different levels of perceived usefulness. The effect size was negligible ($\beta_{\text{usefulness}} = 0.00$), and the posterior probability of a meaningful relationship was 50.9%. This finding suggests that perceived utility does not necessarily translate into greater user engagement, highlighting the complex interplay of factors influencing user behavior with AI tools.

We also examined whether higher acceptance of chatbot technology was associated with greater perceived usefulness. Technology acceptance scores, based on averages of the derived indices from the responses to the Technology Acceptance Questionnaire [2], were modeled as predictors of usefulness ratings. Bayesian cumulative ordinal regression revealed a modest positive relationship, as shown in Fig. 2 [4]. Participants with higher acceptance indices were more likely to rate the chatbot as useful, with increased probabilities of selecting "Agree" or "Strongly Agree." However, the small effect size ($\beta_{\text{acceptance}} = 0.54$, 95% CI : -1.25, 2.24) indicates weak

evidence for a strong association, suggesting that while acceptance contributes to perceptions of usefulness, its practical significance may be limited.

Turning to help-seeking behavior, we hypothesized that individuals with higher help-seeking tendencies would interact more frequently with the chatbot. Help-seeking scores were derived from responses to the adapted Computer Science Help-Seeking Scales [22], while interaction counts were measured as the total number of chat interactions logged during the study sessions. Bayesian Gaussian regression revealed a slight positive association between help-seeking indices and interaction counts ($\beta_{\text{help-seeking}} = 0.33$), but Fig. 2 [5] shows a flat relationship overall, with wide credible intervals indicating weak evidence for this hypothesis.

Interestingly, while help-seeking behavior did not predict interaction frequency, it strongly correlated with technology acceptance. As shown in Fig. 2 [6], participants with higher help-seeking indices had significantly higher predicted acceptance scores ($\beta_{\text{help-seeking}} = 0.46$, 95% CI : 0.26, 0.66), with a posterior probability of 100%. This highlights the role of individual differences in shaping attitudes toward AI tools.

The role of metacognitive awareness, hypothesized to predict help-seeking tendencies, was also investigated. Metacognitive awareness scores were derived from responses to the Metacognitive Awareness Listening Questionnaire [30]. Bayesian regression confirmed a strong positive relationship, as shown in Fig. 2 [7]. Participants with higher metacognitive awareness were significantly more likely to engage in help-seeking behavior ($\beta_{\text{metacognition}} = 0.76$, 95% CI : 0.15, 1.36), with a posterior probability of 99.25%. This underscores the importance of cognitive strategies in optimizing interactions with AI-driven educational tools.

Lastly, we examined whether interaction frequency predicted performance outcomes. Performance scores were calculated as the total number of correctly answered quiz questions. Bayesian linear regression revealed no meaningful relationship, as shown in Fig. 2 [8]. The effect size was negligible ($\beta_{\text{interactions}} = 0.00$), and the posterior probability of a positive effect was 50%, providing no substantial evidence for this hypothesis. This finding suggests that factors beyond interaction frequency, such as individual differences and prior knowledge, play a more significant role in performance. The absence of a strong correlation raises concerns about whether chatbots primarily function as engagement tools rather than as effective learning aids. It is possible that chatbot interactions alone do not sufficiently reinforce learning or promote the deeper cognitive processing necessary for knowledge retention. Future research should explore ways to integrate chatbot interactions with guided metacognitive strategies to enhance their educational impact.

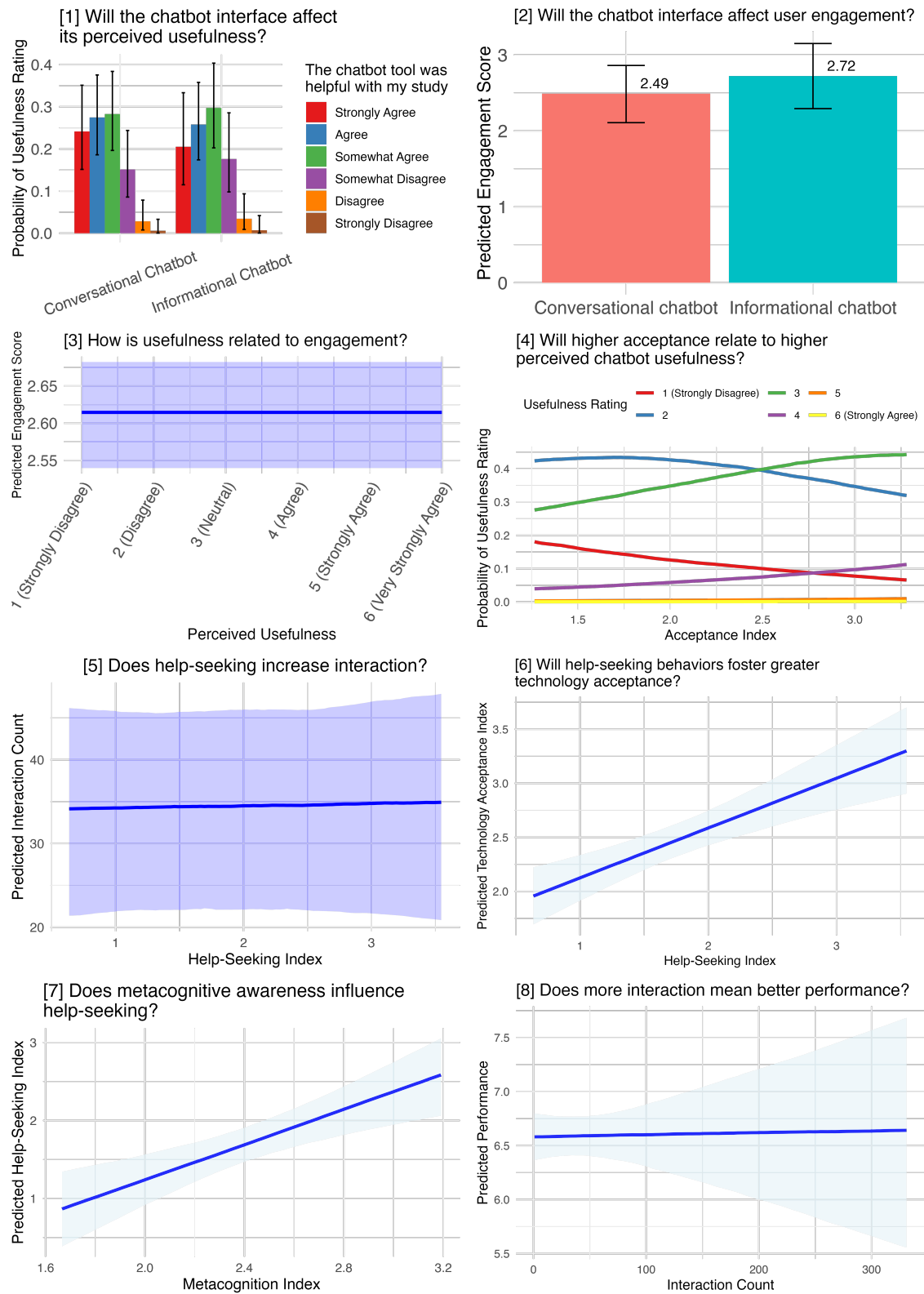


Figure 2: Conditional effects plots showing the relationship between target variables and predicted probabilities.

The findings underscore the nuanced role of chatbot design in fostering engagement and user acceptance. Informational chatbots appear to promote slightly higher engagement, potentially due to their straightforward interaction style, whereas conversational interfaces may be more beneficial in contexts requiring deeper engagement. These observations align with research indicating that overly complex interfaces can induce cognitive overload, thereby diminishing their effectiveness in educational settings [14, 15]. These results suggest that designers should carefully consider the cognitive load associated with different chatbot styles, ensuring that AI-driven educational tools provide structured yet flexible interactions to sustain engagement without overwhelming users. Moreover, the limited effect sizes indicate that chatbot design alone may not be sufficient to drive meaningful engagement. Instead, complementary instructional strategies—such as scaffolding and guided exploration—should be integrated to enhance learning outcomes. Future chatbot iterations should also explore adaptive features that personalize interactions based on users' metacognitive awareness and learning preferences. By tailoring responses to individual needs, chatbots could enhance both engagement and educational effectiveness, ultimately providing a more supportive learning environment.

4 Conclusion and Future Work

This study examined how different chatbot interfaces influence ESL learners' engagement, perceived usefulness, and performance during a learning task. The findings highlight the intricate interplay between interface design, user behavior, and perceived outcomes. While chatbot type (informational or conversational) had minimal practical impact on perceived usefulness or engagement, the slight advantage observed for informational chatbots in these metrics suggests that simple, task-focused designs may be more effective in certain contexts. However, the small effect sizes indicate that this advantage is not definitive, necessitating further research to determine whether specific chatbot design modifications could lead to more substantial improvements in engagement and learning outcomes. The negligible relationship between perceived usefulness and engagement underscores the complexity of user interactions with AI-driven educational technologies, suggesting that additional factors influence how learners engage with these tools. Future studies should explore variations in chatbot interactivity, such as adaptive responses based on learner proficiency, to determine whether deeper engagement can be fostered through dynamic, rather than static, conversational strategies. By refining chatbot adaptability and responsiveness, future iterations may offer more effective support for diverse learning needs.

Help-seeking behavior emerged as a significant predictor of technology acceptance, with individuals more inclined to seek help viewing AI tools more favorably. However, no substantial evidence linked help-seeking behavior to increased interaction counts, suggesting that favorable perceptions do not necessarily translate into frequent tool usage. The critical role of metacognitive awareness in driving help-seeking behavior was also confirmed, reinforcing its importance in AI-based educational interventions. Although the Bayesian modeling framework provided robust insights despite the small sample size, the wide credible intervals in several analyses emphasize the need for caution in interpreting these results. Larger

samples in future research will be necessary to validate and extend these findings.

From a design perspective, the findings highlight the need to balance trust and autonomy in AI-driven educational systems. To enhance engagement, chatbot developers should implement adaptive response mechanisms that adjust based on user confidence levels and learning behaviors (e.g., [31]). For instance, incorporating interactive feedback loops—where the chatbot asks clarifying questions or suggests tailored study strategies—could help sustain engagement while reinforcing learning objectives. Furthermore, personalizing chatbot interactions by integrating user performance data could enable more context-aware responses, ensuring that engagement leads to meaningful learning improvements. These developments would require further research to test their efficacy. For instance, excessive reliance on AI tools may reduce opportunities for critical thinking and independent learning. Striking a balance between guided assistance and learner autonomy is essential to maximizing the pedagogical value of these systems. Equally important is fostering positive user attitudes, as sustained engagement depends on learners perceiving these tools as both effective and trustworthy. As AI technologies continue to shape the educational landscape, understanding their role in promoting equitable and effective learning experiences remains critical to addressing the diverse needs of learners.

This work opens several avenues for further investigation. First, refining the methodology based on reviewer feedback will help address current limitations and improve the study's rigor and scope. Second, future studies should explicitly model the role of ability factors, including topic familiarity, confidence, and language competency, in influencing performance outcomes. Third, exploratory analyses such as topic modeling, sentiment analysis, and sequence analysis could uncover latent themes in user interactions and reveal patterns of behavior over time. Fourth, open-ended survey questions could be used to gather qualitative feedback on user experiences, including perceived challenges and preferences when interacting with chatbots. Finally, increasing the participant pool will enhance the generalizability of findings and provide greater statistical power for hypothesis testing and validating preliminary results. These efforts will deepen our understanding of how AI tools can be optimized to meet the diverse needs of learners across various educational contexts.

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